

Where is our AI Leadership (not followship)

01 / OUTLOOK

The missing white Paper

Setting the agenda with "The Future of AI", reviewed and backed by our advisory board.

02 / PURPOSE

The missing web presence

Answering the definitive question: "Why study CS/SE in this age of AI?"

- Work in progress/ much work on branding.

03 / STRATEGY

Where is our AI Strategy?

Applying CS/SE theory to long-term issues. Built to outlast volatile model lifecycles.

Fuel to acquire more resources, more staff line items?

04 / TACTICS

Where are our AI Tactics?

Sharing practices via regular **lunch swap shops** to continuously refine our skills and digital assets.

Problems with Industrial AI?



01 / ECONOMICS

Unsustainable Costs

Is there an AI bubble collapse incoming? There is an urgent need to mix and match expensive and cheap methods to survive.



02 / RELIABILITY

Poor Testing

Many software bugs take months or years to emerge. Maintainability issues and security attack vectors remain major, unaddressed threats.



03 / EDUCATION

Workforce Deskilling

Where will tomorrow's senior engineers come from? Promoted by the false perception that $CS = SE =$ bracket matching.



04 / MANAGEMENT

New Challenges

Financial issues. Navigating the gap between AI hype and true productivity. Unclear compliance standards and data privacy risks. Copyright ambiguity and intellectual property nightmares.

Solutions to Problems with Industrial AI

01 / ECONOMICS

Hybrid AI: neurosymbolics

LLMs for dialog & interface, with faster, cheaper methods "below the waterline" (Bayesian optimization [1], text retrieval [2]).

02 / RELIABILITY

More Testing, of everything, always

Focus on testing, critical thinking, evaluation, experimental methods, and statistics. Out-of-box testing & crystal clear domain definitions.

03 / EDUCATION

Active Learning

Train lifelong learners. Combine highly structured projects (fewest key abstractions) with unstructured, open-ended problems.

04 / MANAGEMENT

Process & Theory

Requirements engineering (stakeholder trade-offs, MVP, release planning), management theory, economics, and business cases.

[1] <https://arxiv.org/pdf/2607.02583>

[2] <https://arxiv.org/pdf/2606.03640>

Some Examples: Requirements Engineering

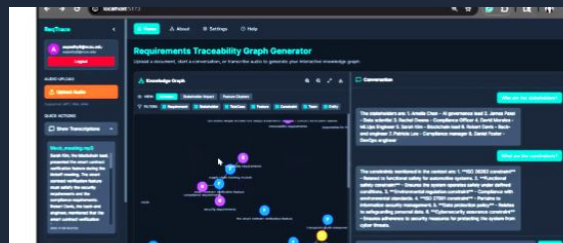
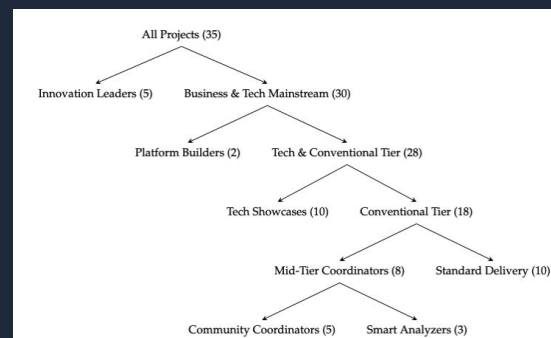
CS510 / COURSE PROJECT

Food Delivery App

Design a food delivery app.
Make it different to every other app out there.

Use LLMs to explore requirements, health and legal documents, etc.

Level	Customer Type	Value Proposition
level=1 (lowest)	One customer type, one value proposition.	Individual. One person at a time, no network effects.
level=2	Multi-Sided. Multiple customer segments, different value props.	Community. Neighborhood, campus, or small group benefits.
level=3	Platform (Early). Enables limited 3rd-party participation, has APIs.	Institutional. Organization-wide or city-level systems.
level=4	Platform (Mature). Robust ecosystem, 3rd parties build businesses.	Societal. Regional, industry-wide impact, potential policy influence.
level=5 (highest)	Infrastructure. Industry foundational layer, platforms built on it.	Systemic. International transformation, new infrastructure.
Might be used by	Entrepreneurs, product	Social entrepreneurs, policy



The auth_users table is completely **locally**.

High confidence (80%) Medium confidence (50-80%) Low confidence (1-50%) Confident

Click on highlighted items to clarify them.

Edge Case Analysis

6 edge cases generated

- ☒ Enter a valid password but provide an incorrect MFA code multiple times to test account lockout mechanisms.
- ☒ Test login with a password containing special characters and verify proper escaping/handling.
- ☒ Attempt authentication during a server outage using the same credentials and observe system handling.
- ☐ Test system behavior when the network connection is lost immediately after entering the password but before MFA is completed.
- ☐ Try to authenticate using a password and MFA code while the database is under heavy load to assess performance.
- ☐ Attempt to access the auth_users table directly without proper authentication to verify security measures.

Gaming group right now very teaching focus.

- Programmatic initiatives(led by collin + dave + sarah + justin (ish))
 - New applied ai intro sequence csc 202,301 (general to eng)
 - Developing an applied AI minor (hopefully, a transient before the major comes on line)
 - Dave:
 - BIG thing: bachelors in AI (complicated and political : status "on hold". Now reconfiguration to be joint ECE ISE)
 - Collin: master s level MS+AI. Seems to be widespread approval
 - Justin: Now robotics is an issue (much discussion)
- Assessment: dave should present
- Dave: infrastructure for local hosted models
- Is AI policy being set by people not so skilled at AI: all NCSU user all think they are the owners and how to innovate. So how to tell them cs is relevant and critical.
- Everyone "owns" AI. so no one is the expert.

Network systems manager

- Network is real frontier of quantum computing: [Ruozhou Yu: quantum networking](#) Associate Professor, 919-515-7938, ryu5@ncsu.edu
- Cyberphysical , robotics networkking
- What to do? ??? rebrand, what is arc: AI big questions, not currently satisfied. "What AI"?
 - ⇒ throw to everyone to "do you own it?" "what technology re you talking about" "who do you test it?"
 - What is it? How do you know it works?
 - "Ai poorly defined", poorly managed.
 - We know things that are missing now... how to prepare our people to apply those currently missing bits.

Dominic: how ai impacts developers in security. Diversity in the panel. "Ai will turn SE into actual engineers".

Will: pressure test systems, security. Need to get fundamentals write. If we don't teach fundamentals of security. "We have not patch our way out of this". E.g. defense debt. How to protect systems. Detection. Recovery methods. Different architectures that make these things identity and access management. We've been taking shortcuts with these things for years, now it's coming back to bite us.

⇒ management change? Pat morrison, trying to figure out to change learning objectives to include ai.

Why cs should be home for AI: CS has been automating things for 50 years. Why is this any different?